

CodeWeaver comparison with AlphaTrans

Four of AlphaTrans's ten exact Java projects were already measured in the published CodeWeaver matrix. All 12 translations compiled; none passed every available fixed runtime case. Across repetitions, 1130/3543 fixed cases passed. AlphaTrans's fragment-level TPR is preserved separately because it is not the same unit or denominator as CodeWeaver's project-test execution.

Scope

The exact common subjects are commons-cli, commons-csv, commons-fileupload, and commons-validator. Six paper subjects were not run, so this artifact makes no ten-project aggregate claim.

Unavailable surface

AlphaTrans's manual type-map completion, manual repair effort, GraalVM fragment validation, and human bug taxonomy cannot be recovered from CodeWeaver's end-to-end project outputs and are reported as unavailable rather than zero.

Exact-subject comparison

Project	AlphaTrans syntax	AlphaTrans TPR	CW builds	CW fixed tests
commons-cli	100.00%	10.08%	3/3	1103/1143 (96.50%)
commons-csv	98.72%	0.00%	3/3	0/894 (0.00%)
commons-fileupload	100.00%	63.44%	3/3	27/117 (23.08%)
commons-validator	99.23%	11.70%	3/3	0/1389 (0.00%)

CodeWeaver repetitions

Rep	Build	Pass all	Fixed tests	Test rate
1	4/4	0/4	379/1181	32.09%
2	4/4	0/4	373/1181	31.58%
3	4/4	0/4	378/1181	32.01%

CodeWeaver execution and model-use telemetry

Scope	Cells	Elapsed h	Assistant turns	Tool calls	Premium requests	AIU	Output tokens	Input tokens
repetition 1	4	16.93	5325	9273	33	29125.47	3376395	N/A (unavailable)
repetition 2	4	16.37	5253	9069	0	28834.56	3194237	N/A (unavailable)
repetition 3	4	14.51	5203	8791	0	26315.46	3188225	N/A (unavailable)

Scope	Cells	Elapsed h	Assistant turns	Tool calls	Premium requests	AIU	Output tokens	Input tokens
all measured cells	12	47.81	15781	27133	33	84275.49	9758857	N/A (unavailable)

CodeWeaver standardized coverage measurements

Rep	Metric	Cells	Mean	Min	Max
1	coverage_before	4	54.06%	38.33%	90.71%
1	standardized_coverage_before	4	54.06%	38.33%	90.71%
1	standardized_coverage_after	4	68.48%	51.69%	90.71%
2	coverage_before	4	50.76%	35.26%	88.76%
2	standardized_coverage_before	4	50.76%	35.26%	88.76%
2	standardized_coverage_after	4	60.95%	35.26%	89.41%
3	coverage_before	4	52.61%	26.58%	96.11%
3	standardized_coverage_before	4	52.61%	26.58%	96.11%
3	standardized_coverage_after	4	69.50%	45.02%	96.11%

Complete source-paper surface audit

Surface	Denominator	Metrics	Artifact status
Table 1	10 projects / 1,797 types	classes, methods, tests, method coverage, ATR, SV, fragments	structured_reference
Table 2	10 projects / 4,654 application-method fragments	syntax, source coverage, GraalVM outcomes, translated-test outcomes, TPR	structured_reference_summary_and_four_project_comparison
RQ2 human study	4 projects / 2 developers	hours, additions, deletions, bug examples	structured_reference
Figure 6	eligible decomposed tests in 9 projects	selected/fail/success ratios and fragment pass-rate distribution	structured_reference_summary
Table 3	10 projects	coverage, decomposed tests, methods per test, TPR+, ATP+	structured_reference_summary
Table 4	10 projects	GraalVM and translated-test outcomes	structured_reference_summary

Surface	Denominator	Metrics	Artifact status
Table 5	10 projects	syntax, GraalVM, translated tests, functional equivalence, cost	structured_reference_summary
Table 6	782 files	context overflow, syntax errors, Graal success, TPR	structured_reference_summary
Figure 7	application-method fragments	DeepSeek-Coder and GPT-4o overlap	structured_reference_summary